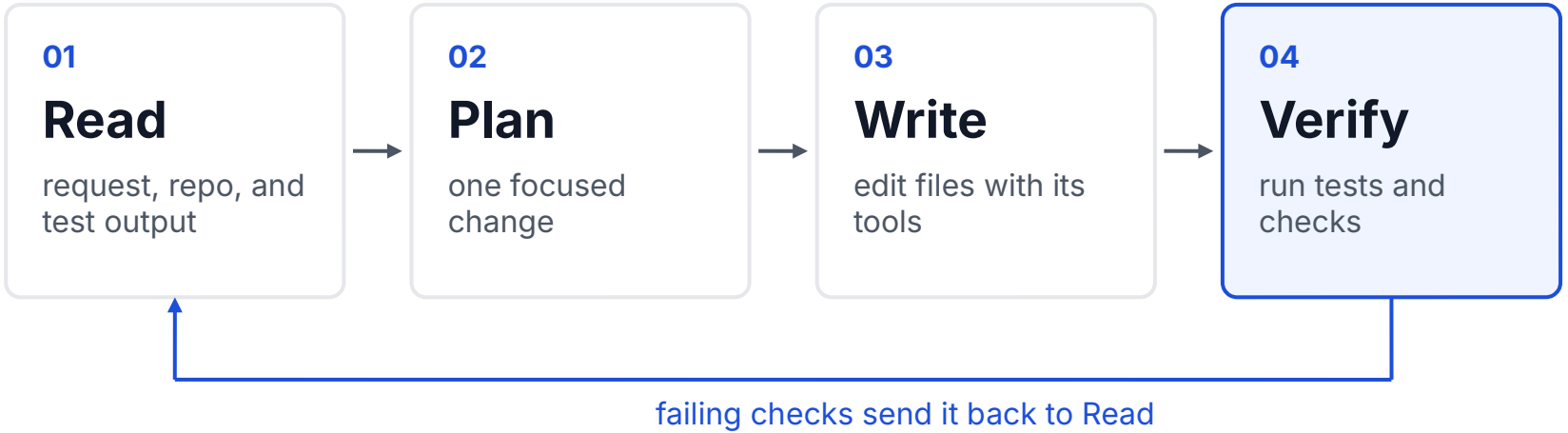
AGENTIC CODING

Agentic coding is a loop, not a single answer

How coding agents read, plan, write, and verify their own work

The agent reads, plans, writes, and verifies in a repeating loop



Tools connect the model to files, commands, search, and tests

`read_file`	see the code
`grep_search`	find what matters
`write_file`	change it
`run_command`	build and test

Without tools, a model can only suggest code. With them, it can check whether the change works.

Tool names are illustrative. Real agents expose different interfaces, permissions, and sandboxes.

Feedback lets each retry build on the last, beyond a 1.96% baseline

1.96%

of 2,294 SWE-bench tasks solved by Claude 2 with BM25 retrieval (2023)

What changes with a loop

- It sees the test output from its last try
- It edits again with that evidence
- Gains vary by model, tools, and task

Source: Jimenez et al., SWE-bench, arXiv:2310.06770v1, Table 5 (2023)

Human review should match each action's consequence, reversibility, and scope

Routine and reversible

Let the agent run

Edits in a branch, tests, local builds

Consequential

Add a checkpoint

Schema changes, dependency upgrades

Irreversible or out of scope

Require approval

Deploys, deletes, anything outside the task